

theFlow!

Product & Technical Overview

Visual Node-Based Canvas Application

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1. What Is theFlow!

theFlow! is a cross-platform desktop application for organising images, video, audio, documents, text, web pages, and PDF files in a freeform infinite canvas. Content is represented as nodes connected by Bézier curves. The result is a visual workspace that feels closer to a physical desk or mood board than a traditional file manager or note-taking app.

The app is written in Python (PyQt6) and distributed as self-contained native binaries for macOS, Windows, and Linux — no installation of Python or any dependency is required on end-user machines.

Key Capabilities

- Infinite 2D canvas with smooth zoom and pan
- 7 node types: Text, Image, Movie, Audio, Document, Paint, Dot
- Inline media viewers that expand below each node (video player, audio waveform)
- Bézier connection curves linking nodes to express relationships
- Backdrop grouping — moves all pinned child nodes together
- Canvas annotation — freehand drawing overlay with color, thickness, and style
- Full undo/redo, copy/paste, import/export of node subsets
- Light and dark themes with live propagation to all canvas items
- .flow file format — JSON, human-readable, forward-compatible
- File association: double-clicking a .flow file opens theFlow! on all platforms

2. Technology Stack

Layer	Technology	Notes
Language	Python 3.9–3.12	3.9 pinned for Linux; 3.12 for macOS/Windows
UI Framework	PyQt6 (Qt6 bindings)	QGraphicsScene/View for canvas; Qt Multimedia for audio/video
Media Playback	Qt Multimedia + FFmpeg	Bundled FFmpeg backend; AVFoundation on macOS
Packaging	PyInstaller 6.x	Freezes Python + Qt into single binary per platform
macOS Installer	DMG (hdiutil)	Drag-to-Applications bundle
Windows Installer	Inno Setup 6	Standard .exe installer with registry file association
Linux DEB	dpkg-deb	.deb package (Ubuntu/Debian)
Linux RPM	rpmbuild	.rpm package (Fedora/RHEL)
Linux AppImage	appimagetool	Portable single-file, runs on any x86_64 Linux (glibc 2.31+)
Persistence	JSON (.flow files)	Human-readable; base64 for embedded binary data

3. Architecture

theFlow! has no package hierarchy — all modules are flat Python files in the project root. The main structural components are:

Module	Responsibility
main.py	App entry point. Platform-specific QApplication subclass, settings load, file-open event routing
view_logic.py	QGraphicsView subclass. All user input: pan/zoom, keyboard shortcuts, node creation, file I/O
scene_logic.py	QGraphicsScene subclass. Serialisation (to_dict/from_dict), undo/redo stack, backdrop child management
node.py	All content node types and their inline media viewers
curve.py	Bézier ConnectionLine and Socket
backdrop.py / note.py	Backdrop grouping and Sticky Note with resize handles
paint_on_canvas.py	Canvas freehand annotation (CanvasPainter, CanvasStroke, StrokeToolbar)
menu.py / ui_components.py	Right-click context menu and Tab creation menu
settings.py / config.py / utils.py	Settings dialog, global constants, shared helpers

4. Build & Distribution

Platform	Command	Output
macOS	<code>./build_mac.sh</code>	theFlow.app + theFlow-0.1.0.dmg
Windows	<code>build_windows.bat</code>	theFlow.exe + theFlow-0.1.0-setup.exe
Linux DEB	<code>./build_deb.sh</code>	theflow + theflow_0.1.0_amd64.deb
Linux RPM	<code>./build_rpm.sh</code>	theflow + theflow-0.1.0-1.x86_64.rpm
Linux AppImage	<code>./build_appimage.sh</code>	theflow + theFlow-0.1.0-x86_64.AppImage

Linux AppImage

The AppImage is a portable, self-contained single-file format that runs on any x86_64 Linux distribution with glibc 2.31 or newer (Ubuntu 20.04+, Fedora 33+, Debian 11+). No installation is required — the file is executable directly:

- Build: `./build_appimage.sh` (from Ubuntu 20.04 or newer)
- Run: `chmod +x theFlow-0.1.0-x86_64.AppImage && ./theFlow-0.1.0-x86_64.AppImage`
- Bundles: Python runtime, PyQt6, Qt6, FFmpeg, ffmpeg binary for audio waveforms
- Settings: stored at `~/.config/theflow/settings.json` (XDG standard)
- Limitation: file association and MIME type registration require manual setup or a companion package

5. File Format

Workspaces are saved as `.flow` files — plain JSON with a predictable structure. Binary content (paint strokes, embedded images) is base64-encoded inline. The format is forward-compatible: unknown keys are ignored on load.

6. Settings

Settings are stored as JSON at a platform-specific path:

- macOS / Windows: next to the executable
- Linux (frozen): `~/.config/theflow/settings.json` (XDG standard)
- Linux (source): `settings/settings.json` in the project root

7. License

theFlow! is free software released under the GNU General Public License v3.0 (GPLv3). Source code is available at github.com/theFlowAppX/theFlow.

Contact: theflow!@protonmail.com